QPU Benchmark Modes: Detailed Analysis and Code-to-Equation Verification

Technical Report for `run_qpu_benchmarks_all_scenarios.py`

OQI-UC002-DWave Project

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1 Executive Summary

This report provides a comprehensive analysis of the three Quantum Processing Unit (QPU) modes implemented in the unified benchmark system for crop rotation optimization. All modes implement **Variant B: Hierarchical Multi-Period Rotation** as defined in the main project report.

1. **qpu-native-6-family**: Direct 6-family BQM without aggregation
2. **qpu-hierarchical-aggregated**: 27→6 aggregation with spatial decomposition and refinement
3. **qpu-hybrid-27-food**: Full 27-food variables with 6-family synergy template

Each mode represents a different trade-off between problem fidelity and QPU compatibility.

2 Problem Formulation: Variant B Multi-Period Rotation

All QPU modes solve the **Variant B** formulation from the main project report. This section reproduces the mathematical formulation for direct comparison.

2.1 Decision Variables

Let $Y_{f,c,t}$ be a binary variable:

$$Y_{f,c,t} = \begin{cases} 1 & \text{if crop } c \in \mathcal{C} \text{ is planted on farm } f \in \mathcal{F} \text{ in period } t \in \mathcal{T} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

2.2 Objective Function

The objective is to maximize the total normalized benefit:

$$\max \quad (\text{Benefit} + \text{Temporal} + \text{Spatial} - \text{Penalty} + \text{Diversity}) \tag{2}$$

2.2.1 Base Benefit (Linear)

$$\text{Benefit} = \sum_{f \in \mathcal{F}} \sum_{c \in \mathcal{C}} \sum_{t \in \mathcal{T}} \frac{B_c \cdot L_f}{A_{\text{total}}} \cdot Y_{f,c,t} \tag{3}$$

where B_c is the composite benefit score for crop c, L_f is the land availability for farm f, and $A_{\text{total}} = \sum_f L_f$.

2.2.2 Temporal Synergy (Quadratic)

$$\text{Temporal} = \gamma_{\text{rot}} \sum_{f \in \mathcal{F}} \sum_{t=2}^{T} \sum_{c_1 \in \mathcal{C}} \sum_{c_2 \in \mathcal{C}} \frac{L_f}{A_{\text{total}}} \cdot R_{c_1,c_2} \cdot Y_{f,c_1,t-1} \cdot Y_{f,c_2,t} \tag{4}$$

The rotation matrix $R \in \mathbb{R}^{|\mathcal{C}| \times |\mathcal{C}|}$ is defined as:

$$R_{c_1,c_2} = \begin{cases} -\beta \cdot 1.5 & \text{if } c_1 = c_2 \text{ (monoculture penalty)} \\ \text{Unif}(\beta \cdot 1.2, \beta \cdot 0.3) & \text{with prob. } p_{\text{frust}} \text{ (disease/competition)} \\ \text{Unif}(0.02, 0.20) & \text{otherwise (beneficial rotation)} \end{cases} \tag{5}$$

2.2.3 Spatial Synergy (Quadratic)

$$\text{Spatial} = \gamma_{\text{spat}} \sum_{(f_1,f_2)\in\mathcal{E}} \sum_{t\in\mathcal{T}} \sum_{c_1\in\mathcal{C}} \sum_{c_2\in\mathcal{C}} \frac{S_{c_1,c_2}}{A_{\text{total}}} \cdot Y_{f_1,c_1,t} \cdot Y_{f_2,c_2,t} \tag{6}$$

where $\mathcal{E}$ is the edge set of the k-nearest neighbor graph and $S_{c_1,c_2} = 0.3 \cdot R_{c_1,c_2}$.

2.2.4 One-Hot Penalty (Quadratic)

$$\text{Penalty} = \lambda_{\text{OH}} \sum_{f\in\mathcal{F}} \sum_{t\in\mathcal{T}} \left(\left(\sum_{c\in\mathcal{C}} Y_{f,c,t} \right) - 1 \right)^2 \tag{7}$$

2.2.5 Diversity Bonus (Linear)

$$\text{Diversity} = \lambda_{\text{div}} \sum_{f\in\mathcal{F}} \sum_{c\in\mathcal{C}} \mathbb{I} \left(\sum_{t\in\mathcal{T}} Y_{f,c,t} > 0 \right) \tag{8}$$

2.3 Model Parameters

Table 1: Variant B model parameters (from main report Table 8)

Parameter	**Symbol**	**Default Value**
Negative synergy strength	β	-0.8
Frustration ratio	p_{frust}	0.70
Temporal rotation weight	γ_{rot}	0.2
Spatial coupling strength	γ_{spat}	$0.5\gamma_{\text{rot}} = 0.1$
Spatial dampening	—	0.3 (applied to S)
One-hot penalty	λ_{OH}	3.0
Diversity bonus	λ_{div}	0.15
Planning periods	T	3

2.4 BQM Conversion

For quantum annealing, the objective is negated (since QA minimizes) and expressed in BQM form:

$$E(\mathbf{Y}) = \sum_{(f,c,t)} h_{f,c,t} \cdot Y_{f,c,t} + \sum_{\substack{(f,c,t)\\(f',c',t')}} J_{(f,c,t),(f',c',t')} \cdot Y_{f,c,t} \cdot Y_{f',c',t'} \tag{9}$$

where $h_{f,c,t}$ are linear biases and J encodes quadratic couplings.

3 QPU Mode Implementations

3.1 Mode 1: qpu-native-6-family

3.1.1 Description

Solves the 6-family problem directly on QPU without aggregation. For scenarios with 6 food families, this is the most direct approach.

3.1.2 Linear Bias Construction

From the BQM conversion of the Variant B objective:

$$h_{f,c,t} = -\frac{B_c \cdot L_f}{A_{\text{total}}} - \frac{\lambda_{\text{div}}}{T} - \lambda_{\text{OH}} \tag{10}$$

```
# Linear bias: negative benefit (BQM minimizes)
linear_bias = -benefit * area_frac
# Add diversity bonus
linear_bias -= diversity_bonus / n_periods
# Add one-hot linear part
linear_bias -= one_hot_penalty

bqm.add_variable(var_name, linear_bias)
```

Listing 1: Linear Bias Implementation

3.1.3 Temporal Synergy Coupling

Implements the Temporal term from Variant B:

$$J^{\text{temp}}_{(f,c_1,t-1),(f,c_2,t)} = -\gamma_{\text{rot}} \cdot R_{c_1,c_2} \cdot \frac{L_f}{A_{\text{total}}} \tag{11}$$

```
# Temporal synergies
for f_idx, farm in enumerate(farm_names):
    area_frac = land_availability[farm] / total_area
    for t in range(2, n_periods + 1):
        for c1_idx in range(n_foods):
            for c2_idx in range(n_foods):
                synergy = R[c1_idx, c2_idx]
                var1 = var_map[(f_idx, c1_idx, t-1)]
                var2 = var_map[(f_idx, c2_idx, t)]
                bqm.add_quadratic(var1, var2,
                    -rotation_gamma * synergy * area_frac)
```

Listing 2: Temporal Synergy Implementation

3.1.4 Spatial Synergy Coupling

Implements the Spatial term with dampened matrix $S = 0.3 \cdot R$:

$$J^{\text{spat}}_{(f_1,c_1,t),(f_2,c_2,t)} = -\gamma_{\text{spat}} \cdot S_{c_1,c_2} = -0.5\gamma_{\text{rot}} \cdot 0.3R_{c_1,c_2} \tag{12}$$

```
# Spatial synergies (within adjacent farms)
for f_idx in range(n_farms - 1):
    for t in range(1, n_periods + 1):
        for c1_idx in range(n_foods):
            for c2_idx in range(n_foods):
                synergy = R[c1_idx, c2_idx] * 0.3  # S = 0.3*R
                var1 = var_map[(f_idx, c1_idx, t)]
                var2 = var_map[(f_idx + 1, c2_idx, t)]
                bqm.add_quadratic(var1, var2,
                    -rotation_gamma * 0.5 * synergy)
```

Listing 3: Spatial Synergy Implementation

3.1.5 One-Hot Penalty

Expanding the quadratic penalty term:

$$\left(\sum_c Y_{f,c,t} - 1\right)^2 = \sum_c Y_{f,c,t} - 2\sum_c Y_{f,c,t} + 2\sum_{c<c'} Y_{f,c,t}Y_{f,c',t} + 1 \tag{13}$$

The quadratic contribution is:

$$J^{\text{OH}}_{(f,c,t),(f,c',t)} = 2\lambda_{\text{OH}} \quad \text{for } c \neq c' \tag{14}$$

```python
# One-hot penalty (quadratic part)
for f_idx in range(n_farms):
    for t in range(1, n_periods + 1):
        vars_this = [var_map[(f_idx, c, t)] for c in range(n_foods)]
        for i in range(len(vars_this)):
            for j in range(i + 1, len(vars_this)):
                bqm.add_quadratic(vars_this[i], vars_this[j],
                    2 * one_hot_penalty)
```

Listing 4: One-Hot Penalty Implementation

3.1.6 Rotation Matrix Generation

```python
def create_family_rotation_matrix(seed: int = 42) -> np.ndarray:
    np.random.seed(seed)
    n_families = 6
    R = np.zeros((n_families, n_families))

    frustration_ratio = 0.7   # p_frust from Variant B
    negative_strength = -0.8  # beta from Variant B

    for i in range(n_families):
        for j in range(n_families):
            if i == j:
                # Same family: monoculture penalty = -beta * 1.5
                R[i, j] = negative_strength * 1.5  # = -1.2
            elif np.random.random() < frustration_ratio:
                # Disease/competition: Unif(beta*1.2, beta*0.3)
                R[i, j] = np.random.uniform(
                    negative_strength * 1.2,
                    negative_strength * 0.3)
            else:
                # Beneficial rotation: Unif(0.02, 0.20)
                R[i, j] = np.random.uniform(0.02, 0.20)
    return R
```

Listing 5: Family Rotation Matrix (matching Variant B Eq. 10)

3.2 Mode 2: qpu-hierarchical-aggregated

3.2.1 Three-Level Pipeline

1. **Level 1 (Classical)**: Aggregate 27 foods → 6 families
2. **Level 2 (Quantum)**: Solve 6-family BQM with spatial decomposition
3. **Level 3 (Classical)**: Refine back to 27 foods for MIQP scoring

3.2.2 Benefit Aggregation

Benefits are averaged by family:

$$B_g = \frac{1}{|\mathcal{C}_g|} \sum_{c \in \mathcal{C}_g} B_c \tag{15}$$

where $\mathcal{C}_g$ is the set of crops in family g.

3.2.3 Spatial Decomposition

Farms are partitioned into clusters of size k:

$$n_{\text{vars}}^{\text{cluster}} = k \times |\mathcal{C}| \times T = k \times 6 \times 3 = 18k \tag{16}$$

For $k = 9$ (default), each cluster has 162 variables, safely below the QPU limit of $\sim$177.

3.2.4 Boundary Coordination

After each iteration, boundary farms receive soft constraints from neighboring clusters:

$$h_{f,c,t}^{\text{boundary}} \leftarrow h_{f,c,t} - 0.5\gamma_{\text{rot}} \quad \text{if } Y_{f_{\text{neighbor}},c,t}^{\text{prev}} = 1 \tag{17}$$

3.3 Mode 3: qpu-hybrid-27-food

3.3.1 Hybrid Rotation Matrix

Uses full 27-food variable space but derives synergy from 6-family template:

$$R_{c_1,c_2}^{27} = R_{g(c_1),g(c_2)}^{6} + \epsilon_{c_1,c_2} \tag{18}$$

where $g(c)$ maps food c to its family and $\epsilon \sim \mathcal{U}(-0.05, 0.05)$.

3.3.2 Cluster Size Constraints

With 27 foods $\times$ 3 periods = 81 variables per farm:

$$k_{\max} = \left\lfloor \frac{170}{81} \right\rfloor = 2 \text{ farms per cluster} \tag{19}$$

4 Data Sources and Plot Mapping

This section documents which result files are used by `generate_single_plot_pdfs.py`.

4.1 Primary Data Files

Table 2: Data files used by the plotting script

Variable	File Path	Description
`DEFAULT_QPU_HIER_PATH`	`qpu_hier_repaired.json`	QPU hierarchical results (repaired)
`DEFAULT_GUROBI_60S_PATH`	`gurobi_baseline_60s.json`	Gurobi 60s timeout baseline
`DEFAULT_GUROBI_300S_FILE`	`@todo/.../gurobi_timeout_test_*.json`	Gurobi 300s timeout results

4.2 Plot Categories and Data Sources

Table 3: Mapping of plot sections to data sources

Plot Section	Data Source	Plots Generated
Section 1: Scaling	QPU + Gurobi 60s	`scaling_objectives_by_vars/farms`, `scaling_time_comparison`, `scaling_qpu_breakdown_by_vars/farms`
Section 2: Split Formulation	QPU + Gurobi 300s	`split_objectives_by_vars/farms`, `split_optimality_gap_by_vars/farms`, `split_solve_time_by_vars/farms`, `split_speedup_by_vars/farms`, `split_pure_qpu_time_by_vars/farms`, `split_gurobi_mip_gap_by_vars/farms`
Section 3: Gap Analysis	QPU + Gurobi 300s	`gap_absolute_comparison`, `gap_ratio`, `gap_vs_mip_gap`, `gap_6family_scaling`, `gap_27food_scaling`, `gap_summary_table`
Section 4: Violation Impact	Dedicated loader	`violation_rate`, `violation_gap_breakdown`, `violation_adjusted_objective`
Section 5: Deep Dive	Dedicated loader	`deepdive_objective_comparison`, `deepdive_ratio_analysis`, `deepdive_gap_attribution`, `deepdive_scatter`, `deepdive_violation_impact_by_scenario`, `deepdive_summary_findings`
Section 6: QPU Advantage	QPU + Gurobi 300s	`advantage_benefit_comparison`, `advantage_benefit_ratio`, `advantage_time_comparison`, `advantage_violations_vs_benefit`, `advantage_pure_qpu_scaling`, `advantage_summary_table`
Section 7: Benchmarks	Benchmark loader	`benchmark_small_*`, `benchmark_large_*`, `benchmark_comprehensive_*`

4.3 Result Files Analyzed

Three JSON result files from the `qpu-hierarchical-aggregated` mode were analyzed:

1. `qpu_hier_all_6family.json`: Hierarchical mode with `farms_per_cluster=5`
2. `qpu_hier_repaired.json`: Hierarchical mode with repair and `farms_per_cluster=9`
3. `qpu_hierarchical_all.json`: Hierarchical mode with `farms_per_cluster=10`

5 Results Analysis

5.1 Objective Quality Comparison

Table 4: MIQP Objective Values by Configuration

Scenario	Farms	All-6Family	Repaired	All-10
rotation_micro_25	5	-5.50	-4.86	-5.75
rotation_small_50	10	-3.88	-21.79	error
rotation_15farms_6foods	15	-5.10	-26.22	error
rotation_medium_100	20	-24.89	-39.24	error
rotation_25farms_6foods	25	-31.15	-52.67	error
rotation_large_200	50	-37.47	N/A	error
rotation_50farms_6foods	50	N/A	-109.67	error

5.2 Key Findings

5.2.1 Embedding Errors

The `qpu_hierarchical_all.json` file shows systematic embedding failures when `farms_per_cluster=10`:

```
"error_message": "Cannot embed given BQM (size 180),
                  sampler can only handle problems of size 177"
```

This occurs because $10 \times 6 \times 3 = 180$ variables exceeds the D-Wave Clique Sampler's limit.

5.2.2 QPU Time Scaling

Pure QPU time scales linearly with number of clusters:

$$t_{\text{QPU}} \approx 40\text{ms} \times n_{\text{clusters}} \times n_{\text{iterations}} \tag{20}$$

5.2.3 Constraint Violations

Rotation violations are consistently zero after repair, but one-hot violations persist, indicating the penalty $\lambda_{\text{OH}} = 3.0$ may be insufficient.

6 Code-to-Equation Verification Summary

Table 5: Mapping of Variant B equations to implementation

Variant B Term	Mathematical Form	Code Implementation
Base Benefit	$\frac{B_c \cdot L_f}{A_{\text{total}}} \cdot Y_{f,c,t}$	`-benefit * area_frac`
Temporal Synergy	$\gamma_{\text{rot}} \cdot R_{c_1,c_2} \cdot \frac{L_f}{A_{\text{total}}}$	`-rotation_gamma * synergy * area_frac`
Spatial Synergy	$\gamma_{\text{spat}} \cdot S_{c_1,c_2} = 0.5\gamma \cdot 0.3R$	`-rotation_gamma * 0.5 * synergy` where `synergy = R * 0.3`
One-Hot Penalty	$2\lambda_{\text{OH}}$ (quadratic term)	`2 * one_hot_penalty`
Diversity Bonus	λ_{div}/T	`diversity_bonus / n_periods`
Monoculture Penalty	$R_{c,c} = -\beta \cdot 1.5 = -1.2$	`negative_strength * 1.5`

All code implementations correctly match the Variant B formulation from the main project report.

7 Recommendations

7.1 Cluster Size Guidelines

1. For **6-family mode**: Use $k \leq 9$ farms/cluster (162 variables max)
2. For **27-food hybrid**: Use $k = 2$ farms/cluster (162 variables)
3. Add safety margin: target 160 variables to avoid edge cases

7.2 Mode Selection

Table 6: Recommended QPU mode by scenario type

Scenario Type	Recommended Mode	Rationale
Small 6-food (≤10 farms)	qpu-native-6-family	Direct embedding
Medium 6-food (11–200 farms)	qpu-hierarchical-aggregated	Decomposition
Any 27-food	qpu-hierarchical-aggregated	27→6 aggregation
Very large (500+ farms)	qpu-hierarchical-aggregated	Scales via clustering